LESSON 9.1

TYPES OF DATA

LESSON INTRODUCTION

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

LEARNING TARGETS

- I can classify data as categorical or numerical.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- What is categorical data? What is numerical data?
- What types of questions can you ask to collect categorical/numerical data?
- What is a statistical question?

LESSON NARRATIVE

Students will learn to classify statistical data as categorical or quantitative. Data will be presented numerically, in tables, or as graphs. Students will create statistical questions that generate numerical data. Student-gathered data will be used for examples.

COMPONENT SUMMARY

9.1.1 WARM-UP *(~5 MIN)*

Students answer reflective questions related to how to use data

9.1.2 GUIDED INSTRUCTION *(15 MIN)*

Review statistical questions, variability, and classifying data as categorical or numerical

9.1.3 EXPLORATION *(10 MIN)*

Complete a class survey on a variety of questions and then review the class data set

9.1.4 GUIDED PRACTICE *(5 MIN)*

Classify whether data in scenarios is categorical or statistical and practice creating statistical questions

9.1.5 YOUR TURN! *(5 MIN)*

Practice classifying variables as categorical or numerical

9.1.6 WRAP-UP *(~5 MIN)*

Find the error made by a student classifying numerical and categorical data

RESOURCES

GLOSSARY

Key Terms:

- Data
- Statistical question
- Categorical data
- Numerical data

Additional Terms:N/A

MATERIALS

- (Optional) Highlighters
- (Optional) Chart Paper
- (Optional) Markers

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that all data with numbers must be quantitative.

THINGS TO CONSIDER

- Consider giving students some simple examples of surveys before beginning the Warm-Up.
- When describing categorical data, include labels, names, colors, or examples.
- Discuss how data is collected and what it can be used for.
- For 9.1.3 Exploration question 2, share the data in a display for students.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

9.1.2 Guided Instruction provides an opportunity to engage students in a Notice and Wonder routine. Position students to notice the difference between the types of data and draw conclusions about how to classify data sets and variables.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

In this lesson, students are analyzing and classifying types of data. Encourage students to help and support each other through the lesson, specifically 9.1.3 Exploration and 9.1.4 Guided Practice. As students learn to identify statistical data, they will build perseverance thus enabling them to solve more challenging problems.

DIFFERENTIATE INSTRUCTION

This opportunity provides auditory entry points through partner discussion and 9.1.2 Guided Instruction. Consider ways to provide visual entry points for learners to engage with the content, such as bar graphs. It may also be helpful for students to use different colored pencils or highlighters to label the types of data. This will provide a visual for students to make connections between data sets and the types of data.

9.1.1 WARM-UP: USING DATA

Component Overview: *Students create survey questions and determine ways in which survey data can be used.*

TEACHER MOVES

Encourage a variety of question types. Students may adjust their suggestions for questions based on how they think the administration could use the data.

9.1.1 Warm-Up: Using Data

At the end of the school year, the school administration will ask students to participate in a survey about their favorite class.

1. **What are two possible questions the survey could ask?**
   Answers will vary.
   What is your favorite class?
   How much time do you spend on homework?

2. **Makayla thinks it would be helpful to ask students to rank their current classes on a scale of 1-6, with 1 being the students' favorite class and 6 representing their least favorite. Explain how this data could be useful.**
   Answers will vary.
   The data could be useful because it could capture more information than just one class per student.
   Students can compare their preferences with others.

9.1.2 Guided Instruction: Types of Data

A ***statistical question*** is a question that can be answered by collecting data. Often, there will be variability in the data.

Examples and non-examples of statistical questions are shown.

Statistical Questions	NOT Statistical Questions
• How many siblings do the students in my class have? • How tall are the players on the girls' basketball team? • What is the favorite song of the 6th graders at my school?	• What time does school start? • Who is the President of the United States? • What is Avery's favorite color?

1. With your partner, discuss the differences between the two types of questions. What do you notice?
   Statistical questions generate data that has many possible answers. Non-statistical questions have one correct answer.

Data are values that are collected together for reference or analysis.

2. What do you think "variability in the data" means?
   Answers will vary. Variability means there are variations, or differences, in responses.

3. Work together to develop a statistical question. Use your question to gather five data values from your classmates. The answers to your question should all be numeric. Your classmates' responses are the data values.
   a. Complete the table. *Answers will vary.*

Question				
How much time do the students in my class typically spend on homework per night?				
Data				
60 minutes	*25 minutes*	*45 minutes*	*30 minutes*	*60 minutes*

   b. Does the data from your question show a lot of variability or little variability?
   Answers will vary based on question in Part A. Yes, there is a lot of variability.
   c. Explain how you determined your description of the variability.
   The time spent on homework for the student with the longest time is over twice the time spent for the student with the shortest time.

4. Two statistical questions are shown.
   Question 1: "How many siblings do the students at my school have?"
   Question 2: "How many hours do the teachers at my school spend outdoors on a weekend?"

   a. Explain which question you would expect to have more variability.
   Question 2
   b. Discuss your choice with your partner.
   Answers will vary. It is reasonable to think that the students have about the same number of siblings, but some teachers may spend much more time than others outdoors on the weekend.

9.1.2 GUIDED INSTRUCTION: TYPES OF DATA

Component Overview: *In this Guided Instruction, students review the formal vocabulary terms used to describe data sets. Students work in partners to create a statistical question and gather data. Students analyze the data to determine the level of variability and then review the different types of data. Students apply the definitions to classify types of data in a table.*

TEACHER MOVES

Notice and Wonder

Engage the class in a Notice and Wonder routine to determine the differences between the two types of questions. Position students to consider what makes a question statistical (i.e., the questions can have many different responses, there is no right answer, etc.) and draw the conclusion that nonstatistical questions have one correct response.

DIFFERENTIATE INSTRUCTION

For English Language Learners whose home language is Spanish, point out that "variability" in English translates to "variabilidad" in Spanish.

ENRICHMENT

Students may struggle to understand what it means for a data set to be more variable. Ask clarifying questions:

- What might you look for to determine variability in a question?
- Can you make a model or illustration to prove your thinking?

9.1.2 GUIDED INSTRUCTION: TYPES OF DATA (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having students create a Frayer Model with these vocabulary terms with examples and non-examples to solidify conceptual understanding. Students must understand the characteristics of each classification.

TEACHER MOVES

Students may confuse the classifications of data, especially if the categorical data can be quantified or represented numerically, such as with zip codes or phone numbers. For students already familiar with averages, point out that numerical data can be averaged, whereas categorical data with numbers such as zip codes does not have an average.

Ask clarifying questions to help students discern between the two:

- How can you describe numerical data? Categorical data?
- Can categorical data be represented numerically? Why?
- How can you measure numerical data?

Data can fall into two different classifications — categorical or numerical.

Categorical data is a type of data that is divided into groups. Categorical data are qualitative.

Numerical data, or quantitative data, is a type of data that is measurable. It is collected in the form of numbers.

5. In the table, classify the type of data described, or provide an example of data with that classification.

Data Description	Examples	Data Classification
m&m colors	red, blue, green, yellow, brown	*categorical*
number of pets a student has	0, 2, 1, 4, 8, 2, 2, 1	*quantitative*
zip code	*Answers may vary.* 32940, 32605	*categorical*
Answers will vary. favorite season	*Answers will vary. fall, spring, winter, summer*	categorical
Answers will vary. hours of sleep per night	*Answers will vary.* 5, 6, 7, 8, 9	numerical

9.1.3 Exploration: Class Survey

1. Answer the following questions. Write your answers in the table under the number that corresponds to each question.

   a. How many letters are in your first name?

   b. How old are you, in years?

   c. What is your favorite food?

   d. What is your favorite sport?

   e. How many languages do you speak fluently?

   f. What languages do you speak fluently?

Answers will vary.

Your Responses to Each Question					
a.	b.	c.	d.	e.	f.
6	11	*pizza*	*soccer*	2	*English, Spanish*

2. Review the display of the combined class data.

   a. Which data sets are categorical?
   3, 4, 6

   b. Which data sets are numerical?
   1, 2, and 5

9.1.3 EXPLORATION: CLASS SURVEY

Component Overview: *Students answer statistical questions. Once students have answered each question, collect and display the data for the class to see and review as a class. Then students categorize the questions by type of data.*

QUESTIONING

Consider using technology to allow students to respond to the data in real time (example: Google survey, Formative, Kahoot, etc.). This will allow for a more easily available method of data collection and display.

TEACHER MOVES

Notice and Wonder

Position students to analyze the data through a Notice and Wonder routine.

- What do you notice about the data?
- Which questions, if changed, could alter whether the set is numerical or categorical?
- What do you wonder after looking at the data?

9.1.4 GUIDED PRACTICE: CLASSIFYING DATA

Component Overview: *In this Guided Practice, students classify data as either categorical or numerical and explain how they know. Students then create their own statistical questions that will result in categorical data being collected and numerical data being collected.*

TEACHER MOVES

Encourage students to refer to previous components of the lesson or their Frayer Models if they are struggling to classify data.

TEACHER MOVES

Students may misclassify uniform number and grade level as numeric. Uniform number is a form of identification number, and grade level is a category.

TEACHER MOVES

As students create statistical questions for questions 4 and 5, consider walking around the room and choosing a few questions to share with the class. Write the questions on the board at the front of the class and prompt students to classify the resulting data as quantitative or categorical.

9.1.4 Guided Practice: Classifying Data

1. **A city parks department recently collected information about the breeds of dogs visiting a local dog park. They discovered that over 50 breeds of dogs visited the park on Saturday.**

   Determine if the data is categorical or numerical. Explain your thinking.

   The data is categorical because each answer represents a breed of dog, and the responses can be categorized as each breed.

2. **At the beginning of the season, the football coach creates a roster with the players' names, ages, grade levels, and uniform numbers.**

   a. **Which of the variables are categorical?**
   players' names, uniform numbers, grade level

   b. **Which of the variables are quantitative?**
   ages

   c. **For each variable, explain how you determined the type of data.**
   Age generates numerical answers and variability of data. Players' names and uniform numbers are unique to each player. The responses for grade level can also be placed into categories so the data is categorical.

3. **What is the average weight of an elephant, in pounds?**

   a. **What is your guess?**
   Answers will vary. 12,000 pounds.

   b. **Compare your answer with four other students. What do you notice?**
   Answers will vary.

   c. **Is this categorical or numerical data?**
   numerical

4. **Create a statistical question that will result in categorical data being collected.**
   Answers will vary. In what month were the students in my class born? What season do adults in Florida like the most?

5. **Create a statistical question that will result in quantitative data being collected.**
   Answers will vary. How tall are the women in my family? How many hours a week do teenagers in my neighborhood spend on social media?

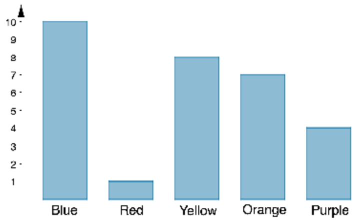
9.1.5 Your Turn!

1. Classify each variable as Categorical (C) or Numerical (N).

   a. Social media followers ___N___
   b. Color of cars in the faculty parking lot ___C___
   c. Shoe size ___N___
   d. Cost of a drink ___N___
   e. Zip code ___C___
   f. Type of pets you have ___C___

2. Will the question, "What is your favorite television or movie streaming service to use?" result in categorical or quantitative data? Explain how you made your choice.
   The responses will create categorical data because the answers will be examples of streaming services, not numbers.

3. Given this bar chart, classify the data as categorical or quantitative.



categorical

9.1.5 YOUR TURN!

Component Overview: *Students work independently to classify variables and data sets as numerical or categorical.*

DIFFERENTIATE INSTRUCTION

Students may struggle to explain their thinking in writing. Consider providing sentence stems to help students with written response:

"The responses will create categorical/numerical data because…"

9.1.6 Wrap-Up: Error Analysis

Bradley's teacher asked him to circle the numerical data in the table below.

Name	Student ID	Age	Hair Color	Grade Level
Rachel	1254866	10	Brown	6
Elijah	2548631	11	Brown	6
Chase	2645852	11	Blonde	6
Ben	2548654	10	Black	6

1. His teacher said that his response was incorrect. Correct his mistake and explain what mistake he made.
   Student ID numbers are unique to each student, so they are categorical data.

9.1.6 WRAP-UP: ERROR ANALYSIS

Component Overview: *Students complete an error analysis and correct a mistake in categorizing data.*

DIFFERENTIATE INSTRUCTION

For students who might struggle with an error analysis, consider scaffolding the question. Ask them first to solve the problem, explain their thinking, and then identify one to two errors a student may have made if they were to attempt a different method.